



Today's agenda

↳ ArrayList internal

↳ Hashmap Construction



AlgoPrep



→ **Arraylist** : **Dynamic Array**

↓
Array

list

10	20	30	40
----	----	----	----

10	20	30	40	50	60	70	80
----	----	----	----	----	----	----	----

0	1	2	3	4	5	6	7
10	20	30	40	50	60	70	80

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

1st add → 10 → $O(1)$

2nd add → 20 → $O(1)$

3rd add → 30 → $O(1)$

4th add → 40 → $O(1)$

5th add → 50 → $O(1)$

6th add → 60 → $O(1)$

7th add → 70 → $O(1)$

8th add → 80 → $O(1)$

9th add → 90 → $O(8+1)$

10th add → $O(1)$

11th add → $O(1)$

12th add → $O(1)$

13th add → $O(1)$

14th add → $O(1)$

15th add → $O(1)$

16th add → $O(1)$

$O(2)$

$O(2)$

$O(2)$

$O(2)$

$O(2)$

→ $O(2)$

→ $O(2)$

→ $O(2)$

→ $O(2)$

→ $O(2)$

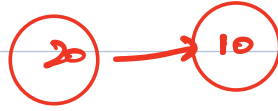
→ $O(2)$

→ $O(2)$

→ $O(2)$



//LinkedList



↳ LinkedList <Integer> ll : new LinkedList<>();

↳ ll.addLast(10); → ll.add(10);

↳ ll.addFirst(20);

```
public class HMNode {  
    int val;  
    String str;  
    HMNode next;  
}
```

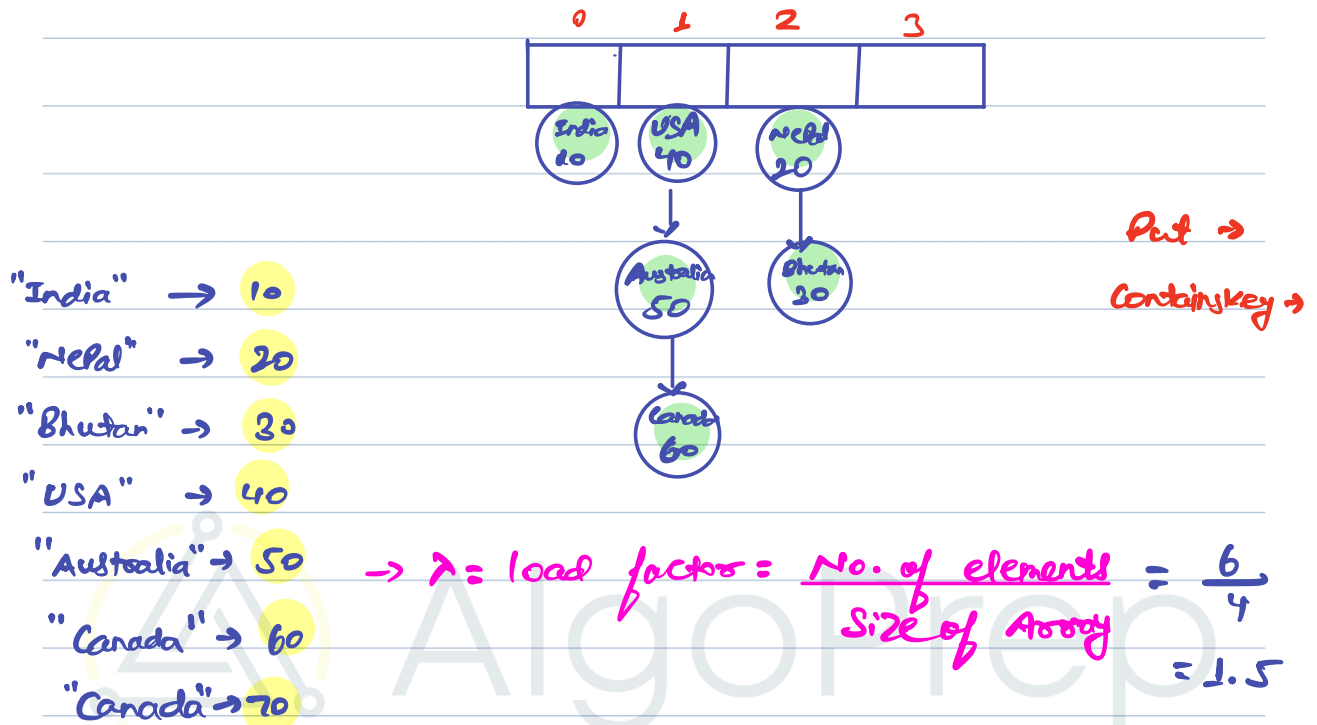
↳ LinkedList <HMNode> ll : new LinkedList<>();

ll.add(new HMNode(10, "Hello"));



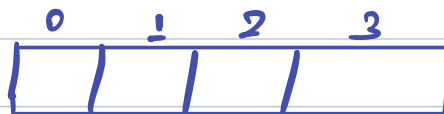


HashMap construction $\rightarrow$ Array of LinkedList.



$\lambda > 2 \rightarrow$ rehash

$\rightarrow$ which index you key belongs $\rightarrow O(1)$



```

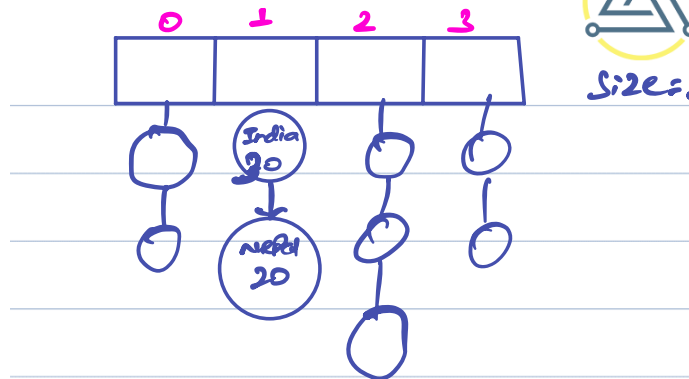
public void put(K key, V val) throws Exception {
    int bi = hashFunction(key);
    int di = findInBucket(bi, key);

    if( di == -1){
        HMNode n = new HMNode(key, val);
        buckets[bi].addLast(n);
        size++;
    }else{
        HMNode n = buckets[bi].get(di);
        n.value = val;
    }

    double lambda = size*1.0/buckets.length;

    if(lambda > 2.0){
        rehash();
    }
}

```



$bi = 1$

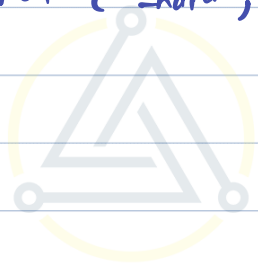
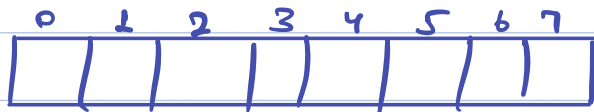
$di = 0$

$$\frac{9}{4} = 2.25$$

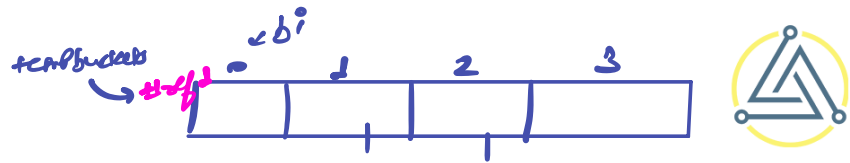
Put ("India", 10)

Put ("Nepal", 20)

Put ("India", 30)

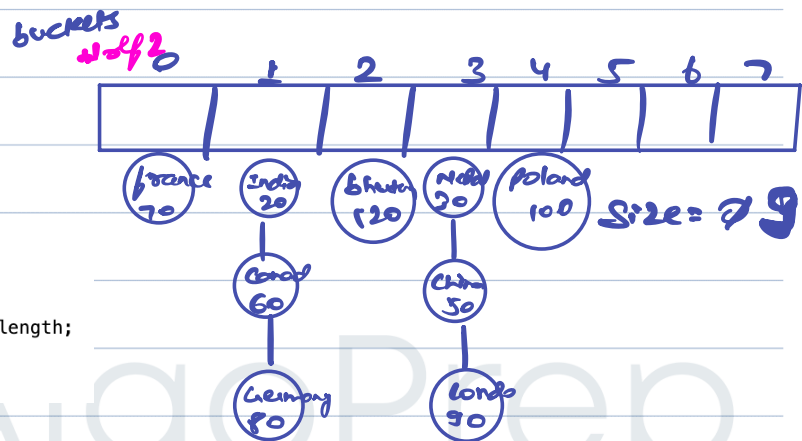


AlgoPrep



```
public void rehash(){
    LinkedList<HMNode>[] tempbuckets = buckets;
    initbuckets(2*tempbuckets.length);
    size = 0;

    for(int bi=0; bi<tempbuckets.length; bi++){
        for(HMNode t: tempbuckets[bi]){
            put(t.key, t.value);
        }
    }
}
```



Bhutan 120

```
public int hashFunction(K key){
    int temp = key.hashCode();
    int bi = Math.abs(temp) % buckets.length;
    return bi;
}
```